



My First Python Adventure!

A Visual Guide to Coding for Beginners

This secret notebook belongs to:

1. Where do we write Code?



To write Python, we use a cool free tool from Google called **Google Colab**. It runs in your browser, so you don't need to install anything!



Steps to Start:

1. Go to colab.research.google.com
2. Click on **"New Notebook"**.
3. Type your code in the gray box (called a "Cell").
4. Click the round **Play Button** to run it!

Here is what it looks like on your screen:



2. Variables & Data

A **Variable** is like a magical digital box. You write a name on the outside, and put information on the inside using an equals sign =.

code:

```
age = 12
```



Types of Data (What's in the box?)

1 2
3 4

Integer (Number)

Whole numbers for math. `score = 100`

abc

String (Text)

Words inside "quotes". `name = "Priya"`



Boolean (True/False)

Like a light switch! `game_over = False`

3. Talking & Listening



Talking: `print()`

Use `print` to make the computer talk or show an answer on the screen.

PYTHON CODE

COLAB CELL

```
print("Welcome to Python!")  
print("I can count to", 3)
```

COMPUTER SCREEN (OUTPUT)

```
Welcome to Python!  
I can count to 3
```



Listening: `input()`

Use `input` to ask the user a question. The computer stops and waits for them to type an answer!

PYTHON CODE

```
color = input("What is your favorite color? ")  
print(color + " is a beautiful color!")
```

COMPUTER SCREEN (OUTPUT)

```
What is your favorite color? Blue  
Blue is a beautiful color!
```

4. The Super Calculator

Python is incredibly fast at math. You can use symbols (operators) to do calculations.

+

Addition

-

Subtraction

*

Multiplication (Star)

/

Division (Slash)

Let's calculate using Variables!

We can store numbers in variables and do math with them.

PYTHON CODE

```
# How many items for the picnic?
sandwiches = 4
apples = 6

total_food = sandwiches + apples
print("Total food items:", total_food)

# If we share with 2 friends:
print("Each friend gets:", total_food / 2)
```

COMPUTER SCREEN (OUTPUT)

```
Total food items: 10
Each friend gets: 5.0
```

5. Making Choices (If / Else)

Computers can make decisions! We give them a condition. **IF** it's true, it goes one way. **ELSE** (otherwise), it goes the other way.

Is it raining? (weather == "rain")

YES (True)



Take Umbrella 🌂

NO (False)



Wear Sunglasses 🕶️

PYTHON CODE (NOTICE THE INDENTATION!)

```
weather = "rain"

if weather == "rain":
    # This runs if True (notice the spaces!)
    print("Take Umbrella 🌂")
else:
    # This runs if False
    print("Wear Sunglasses 🕶️")
```

OUTPUT

Take Umbrella 🌂

6. Lists & Loops

Lists (The Data Train)

A List is like a train where each carriage holds a piece of data. We use square brackets [] to build the train.

toys =

"Lego"

-

"Doll"

-

"Car"

Loops (Doing it again!)

A **For Loop** lets us go through every carriage on our list train, one by one, automatically!

PYTHON CODE

```
toys = ["Lego", "Doll", "Car"]

# For every item inside the toys list:
for item in toys:
    print("I am playing with a", item)
```

OUTPUT

```
I am playing with a Lego
I am playing with a Doll
I am playing with a Car
```

7. Practice Problems



Time to test your superpowers! Try typing these exact codes into Google Colab and watch the magic happen.



Challenge 1: The Pet Name Generator

Use variables and addition (+) to combine words into a funny name.

```
adjective = "Fluffy"  
animal = "Dragon"  
pet_name = adjective + " " + animal  
print("Your new pet is:", pet_name)
```

```
> Your new pet is: Fluffy Dragon
```



Challenge 2: The Magic Age Predictor

Ask the user for their age, and tell them how old they will be in 10 years!

(Note: input() gives a string, so we use int() to turn it back to a number!)

```
age_text = input("How old are you? ")  
age_number = int(age_text) # Magic trick to make it a math  
number!  
future_age = age_number + 10  
print("In 10 years, you will be", future_age)
```

```
> How old are you? 12  
> In 10 years, you will be 22
```


More Challenges!



Challenge 3: The Password Door

Use an if/else block to check if the user knows the secret password.

```
secret_pass = "python123"
guess = input("Enter the password: ")

if guess == secret_pass:
    print("Access Granted! Welcome to the secret club. 🕶")
else:
    print("Access Denied! Intruders alert! 🚨")
```

```
> Enter the password: apple
> Access Denied! Intruders alert! 🚨
```



Challenge 4: The Snack Machine

Use a for loop to print a menu of snacks automatically.

```
snacks = ["Chips", "Cookie", "Apple", "Juice"]

print("--- SNACK MENU ---")
for item in snacks:
    print("We have:", item)
```

```
> --- SNACK MENU ---
> We have: Chips
> We have: Cookie
> We have: Apple
> We have: Juice
```

8. Final Quiz!

1 Which code is a String (Text)?

A) `100` B) `"Hello"` (Correct!) C) `True`

2 What does `input()` do?

A) Prints a message
B) Asks the user a question (Correct!)

3 To make the computer decide things, we use:

A) `if / else` (Correct!) B) `print` C) `trains`



CERTIFICATE OF ACHIEVEMENT

This certifies that

has successfully learned the basics of Python!

Date



Teacher